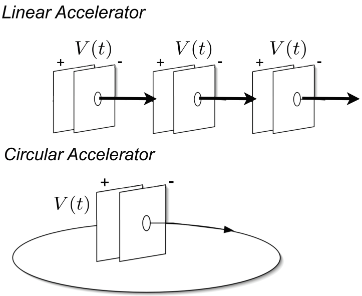
Longitudinal stability

Linear and circular accelerators

[Follow the lecture by M. Syphers, USPAS 2018]

- Linear accelerator:
  - pass N cavities 1 time each
  - can have different frequences/phases
  - typically, one frequency per section
- Circular accelerator:
  - pass 1 cavity N times
  - single frequency that might change if particle speed changes during acceleration (protons/ions)
- Otherwise, essentially the same longitudinal dynamics: accelerating gap + drift
- Must consider time of flight between cavities/passages

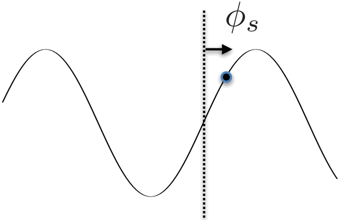


Acceleration

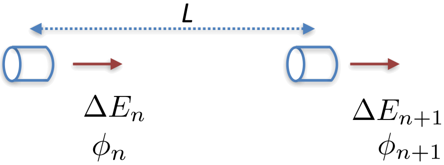
- Consider particles propagating though a system of accelerating cavities
- Each cavity has oscillating fields with frequency f_{RF} and maximum “applied” voltage V (takes into account transition time factor, etc.)
- The ideal particle would arrive at the cavity at phase ϕ_s
- We will choose ϕ_s relative to the “positive zero-crossing” (typical for synchrotrons), then the ideal particle acquires

$$\Delta E_s = \Delta W_s = qV \sin\phi_s$$

- Linacs more often define ϕ_s relative to the “crest” of the RF wave
- Another source of confusion (much like the definition of emittance)



Acceleration II



- Wish to accelerate the ideal particle
- As this particle exits the $(n+1)^{\text{th}}$ RF cavity, we would have

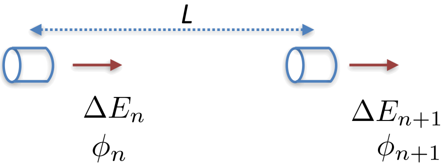
$$E_s^{(n+1)} = E_s^{(n)} + eV\sin\phi_s$$

- For a synchrotron, this would be the total energy gain on the $(n+1)^{\text{th}}$ revolution
- If f_0 is the revolution frequency then the ideal energy gain per second would be

$$\frac{dE_s}{dt} = f_0 eV\sin\phi_s$$

- What is the longitudinal motion near the ideal particle?
- Coordinates: ϕ – phase w.r.t. RF system; $\Delta E = E - E_s$ – energy deviation from the ideal

Difference equations



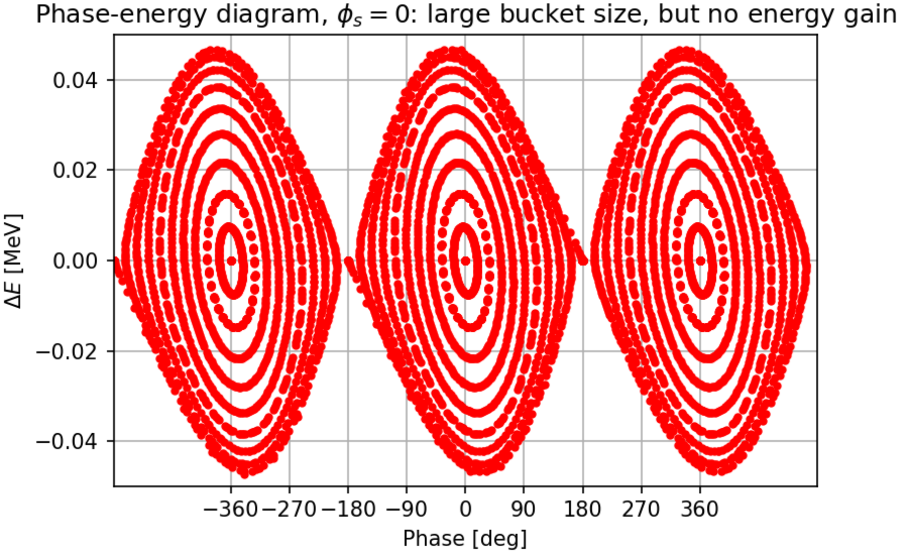
$$\begin{cases} \phi_{n+1} = \phi_n + \dfrac{2\pi h\eta}{\beta^2 E}\Delta E_n, \\ \Delta E_{n+1} = \Delta E_n + eV(\sin\phi_{n+1} - \sin\phi_s), \end{cases}$$

where $h = L/(\beta\lambda)$, $\lambda = c/f_{RF}$ or $h = f_{RF}L/v$.

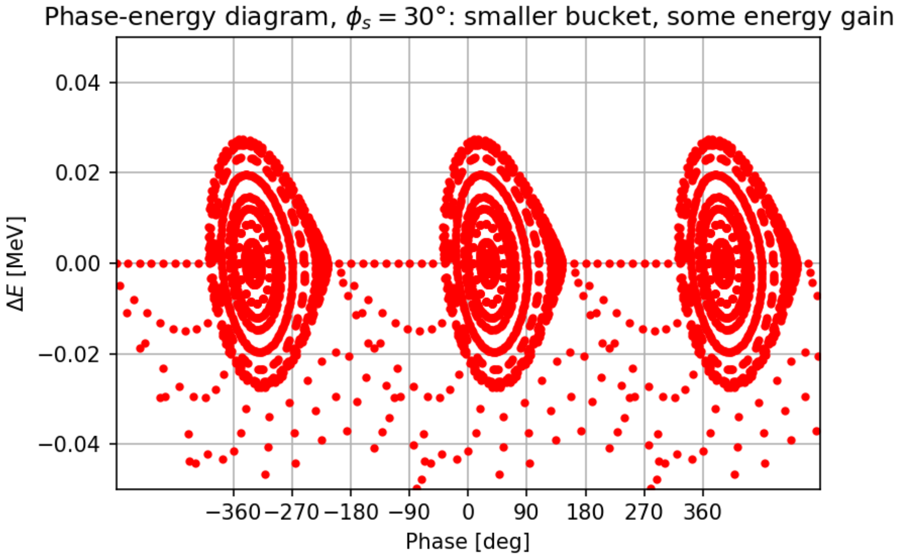
- We want h to be an integer
- If L is the circumference of a synchrotron, $\Rightarrow h = f_{RF}/f_0$, where f_0 is the revolution frequency ($L/v = \tau$, $f_0 = 1/\tau$)
- h is called the "harmonic number"

Let's see how these difference equations work

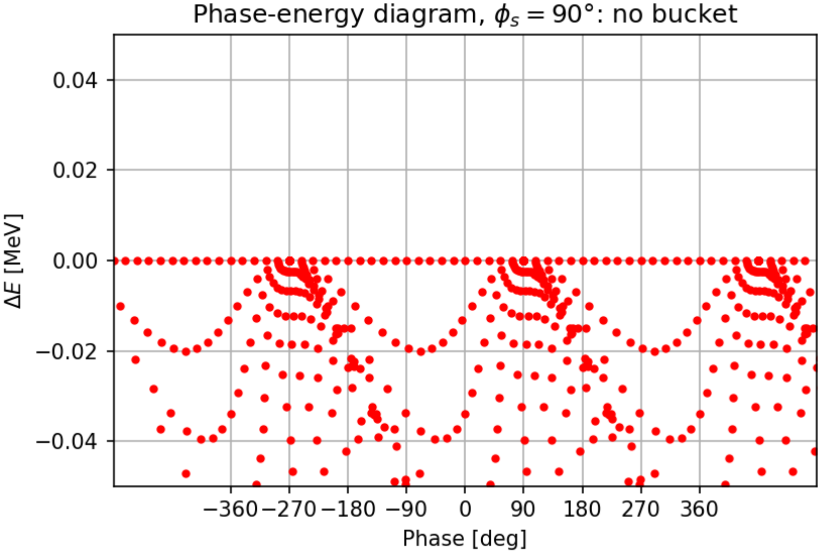
$\phi_s = 0$: no acceleration, largest stable bucket



$\phi_s = 30°$: some acceleration, smaller stable bucket

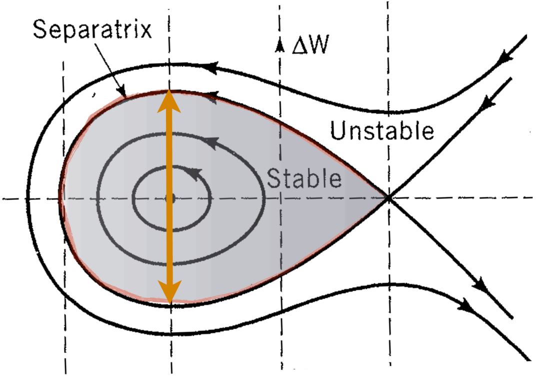


$\phi_s = 90°$: no bucket, only the ref. particle gets accelerated



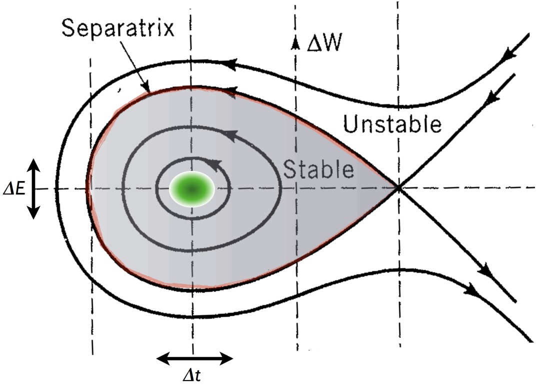
Acceptance and emittance

- Stable region is often called an “RF bucket” (“contains” the particles)
- Maximum vertical extent is the maximum spread in energy that can be accelerated through the system



Acceptance and emittance

- Stable region is often called an “RF bucket” (“contains” the particles)
- Maximum vertical extent is the maximum spread in energy that can be accelerated through the system
- We want the particles to occupy a much smaller area in the phase space



Difference ⇒ differential equation

Start with

$$\begin{cases} \phi_{n+1} = \phi_n + \dfrac{2\pi h\eta}{\beta^2 E}\Delta E_n, \\ \Delta E_{n+1} = \Delta E_n + eV(\sin\phi_{n+1} - \sin\phi_s), \end{cases}$$

Turn into differential equations:

$$\frac{d\phi}{dn} = \frac{2\pi h\eta}{\beta^2 E}\Delta E, \quad \frac{d\Delta E}{dn} = eV(\sin\phi - \sin\phi_s)$$

Hence,

$$\frac{d^2\phi}{dn^2} = \frac{2\pi h\eta}{\beta^2 E}\frac{d\Delta E}{dn} = \frac{2\pi h\eta}{\beta^2 E}eV(\sin\phi - \sin\phi_s)$$

Multiply by $d\phi/dn$, integrate:

$$\int\left(\frac{d^2\phi}{dn^2}\right)\frac{d\phi}{dn}dn - \frac{2\pi h\eta}{\beta^2 E}eV\int(\sin\phi - \sin\phi_s)\frac{d\phi}{dn}dn = 0$$

$$\frac{1}{2}\left(\frac{d\phi}{dn}\right) + \frac{2\pi h\eta}{\beta^2 E}eV(\cos\phi + \phi\sin\phi_s) = \text{constant}$$

Equation of the trajectories in phase space

$$\frac{1}{2}\left(\frac{d\phi}{dn}\right) + \frac{2\pi h\eta}{\beta^2 E} eV(\cos\phi + \phi\sin\phi_s) = \text{constant}$$

Combining with one of the original differential equations

$$\frac{d\phi}{dn} = \frac{2\pi h\eta}{\beta^2 E}\Delta E,$$

we obtain

$$\boxed{\Delta E^2 + 2\frac{\beta^2 E}{2\pi h\eta} eV(\cos\phi + \phi\sin\phi_s) = \text{constant}}$$

This is the equation describing any phase space trajectory we saw earlier in this lecture, just substitute the desired initial conditions.

Synchrotron oscillations

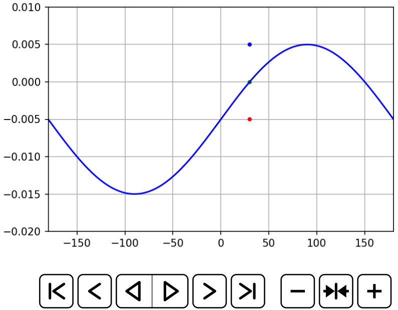
- Particles near the synchronous phase and reference energy will oscillate about the synchronous particle with the "synchrotron frequency"
  - (longitudinal motion is called "synchrotron motion" even in a linac)
- In a synchrotron, the "synchrotron tune" is the number of synchrotron oscillations per revolution
  - (much like the "betatron tune" is the number of betatron oscillation per turn in the transverse direction)
- Take another look at

$$\frac{d^2\phi}{dn^2} - \frac{2\pi h\eta}{\beta^2 E} eV(\sin\phi - \sin\phi_s) = 0$$

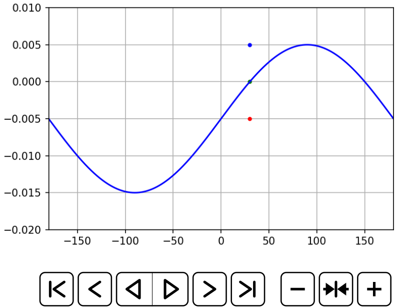
- Let $\phi = \phi_s + \Delta\phi$, $\Rightarrow$
$\sin\phi - \sin\phi_s = \sin\phi_s \cos\Delta\phi + \cos\phi_s \sin\Delta\phi - \sin\phi_s \approx \Delta\phi\cos\phi_s$
-

$$\frac{d^2\Delta\phi}{dn^2} - \left(\frac{2\pi h\eta}{\beta^2 E} eV\cos\phi_s\right)\Delta\phi = 0, \quad \Rightarrow \quad \boxed{\nu_s = \sqrt{-\frac{h\eta eV}{2\pi\beta^2 E}\cos\phi_s}}$$

Phase stability: stable motion around the synchronous phase

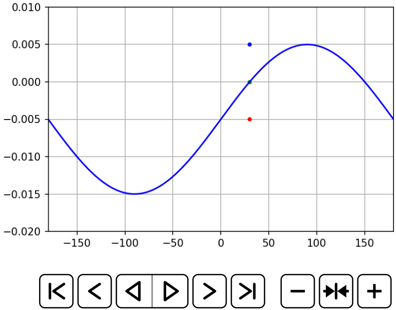


- Slip factor $\eta < 0$ (below transition)
- Phase on the rising slope is stable:



- Slip factor $\eta > 0$ (above transition)
- Phase on the rising slope is unstable:

Phase stability: stable motion around the synchronous phase



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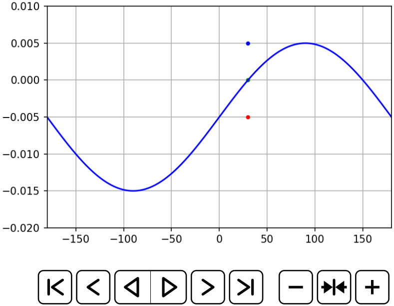

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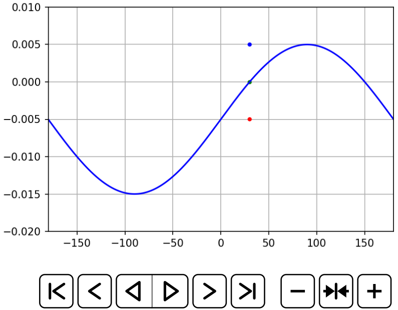

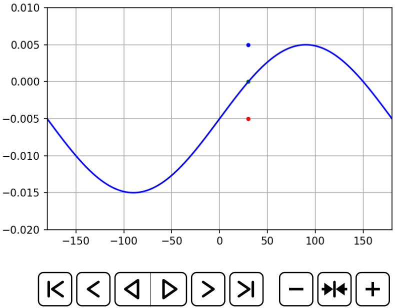

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- Phase on the rising slope is stable:
  - speed changes are more important than path length changes
  - higher momentum particle arrives to the next RF early

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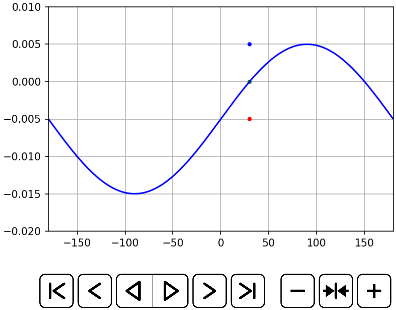

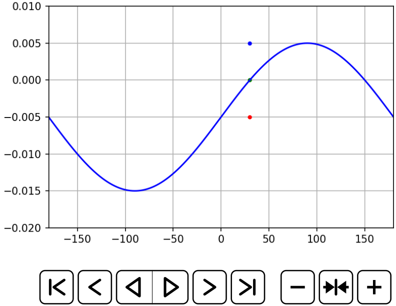

- Slip factor $\eta < 0$ (below transition)
- Phase on the rising slope is stable:
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  - gets smaller kick (less energy)

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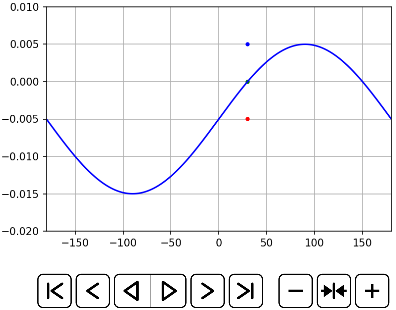

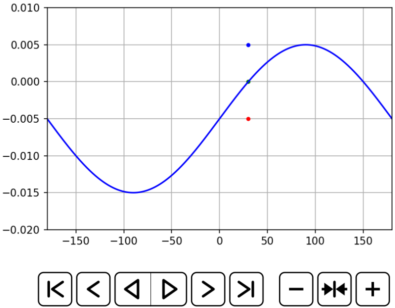

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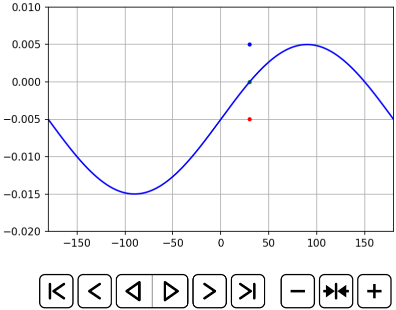

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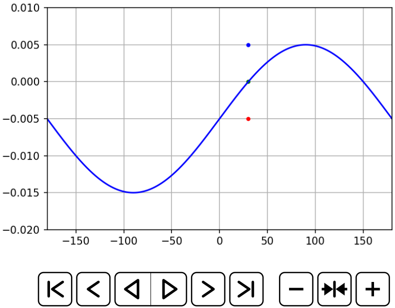

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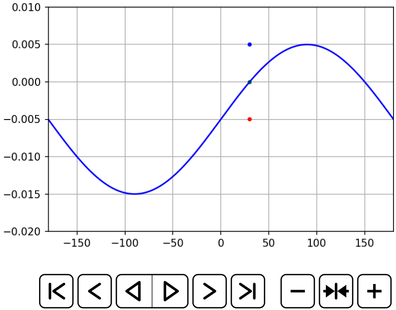

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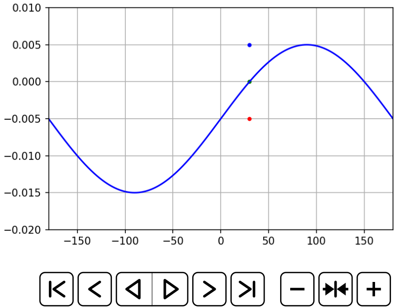

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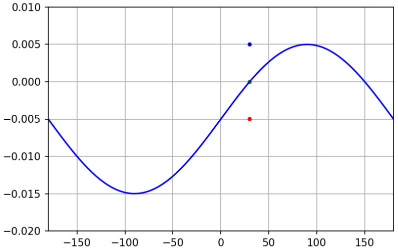

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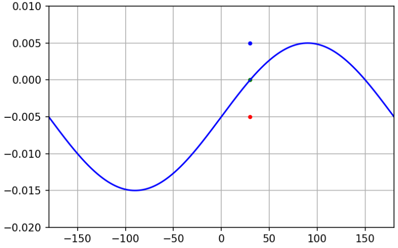

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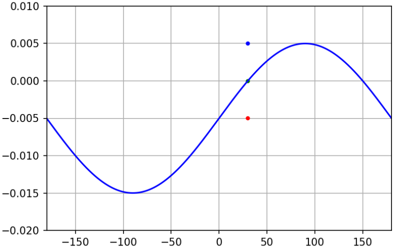

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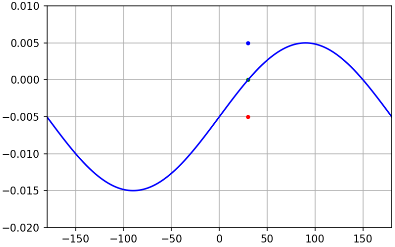


- Slip factor $\eta > 0$ (above transition)
- Phase on the rising slope is unstable:
  - path length changes are more important than speed changes
  - higher momentum particle arrives to the next RF late
  - gets larger kick (more energy)
  - no restoring force

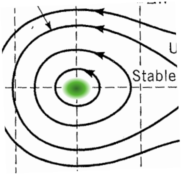
Frequencies of the motion

- From what we've just seen, the synchrotron motion in a circular accelerator takes many (perhaps hundreds of) revolutions to complete one synchrotron period
- On the other hand, in the transverse plane, a particle will typically undergo many betatron oscillations during one revolution
- Thus, transverse/longitudinal dynamics typically occur on very different time scales – this actually justifies us studying them independently

Motion Near the Ideal Particle

Linearize the motion, and write in matrix form…

$$\phi = \phi_s + \Delta\phi$$



$$\begin{aligned}
\phi_{n+1} &= \phi_n + \frac{2\pi h\eta}{\beta^2 E}\Delta E_n \\
\Delta E_{n+1} &= \Delta E_n + QeV(\sin\phi_{n+1} - \sin\phi_s) \\
&= \Delta E_n + QeV(\sin\phi_s \cos\Delta\phi_{n+1} + \sin\Delta\phi_{n+1}\cos\phi_s) - \sin\phi_s) \\
&= \Delta E_n + QeV\cos\phi_s\ \Delta\phi_{n+1} \\
&= \Delta E_n + QeV\cos\phi_s\left[\Delta\phi_n + \frac{2\pi h\eta}{\beta^2 E}\Delta E_n\right]
\end{aligned}$$

Thus,

$$\boxed{\begin{aligned}
\Delta\phi_{n+1} &= \Delta\phi_n + \frac{2\pi h\eta}{\beta^2 E}\Delta E_n \\
\Delta E_{n+1} &= QeV\cos\phi_s\Delta\phi_n + \left(1 + \frac{2\pi h\eta}{\beta^2 E}QeV\cos\phi_s\right)\Delta E_n
\end{aligned}}$$

or,

$$\begin{pmatrix} \Delta\phi \\ \Delta E \end{pmatrix}_{n+1} = \begin{pmatrix} 1 & \frac{2\pi h\eta}{\beta^2 E} \\ QeV\cos\phi_s & \left(1 + \frac{2\pi h\eta}{\beta^2 E} QeV\cos\phi_s\right) \end{pmatrix} \begin{pmatrix} \Delta\phi \\ \Delta E \end{pmatrix}_n$$

$$= \begin{pmatrix} 1 & 0 \\ QeV\cos\phi_s & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{2\pi h\eta}{\beta^2 E} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \Delta\phi \\ \Delta E \end{pmatrix}_n$$

$$M = M_c \cdot M_d$$

"thin" cavity — *drift*

(*acts as longitudinal focusing element*)

Note: for $\eta < 0$, M_d is a "backwards" drift; i.e., $\Delta\phi$ decreases for $\Delta E > 0$

(when no bending)

$\eta = -1/\gamma^2$ in straight region (linac)

Remember from transverse motion, $x \propto \sqrt{\beta} \sin \Delta\psi$
and when *M* was periodic,

$$M = \begin{pmatrix} \cos\Delta\psi + \alpha \sin\Delta\psi & \beta \sin\Delta\psi \\ -\gamma \sin\Delta\psi & \cos\Delta\psi - \alpha \sin\Delta\psi \end{pmatrix} \qquad \text{and} \quad tr M = 2\cos\Delta\psi$$

$\Delta\psi$ = phase advance through periodic section

Can imagine "longitudinal" β, α, γ, $\Delta\psi$ parameters as well
Note: from *M* of previous page, if represents periodic structure (synchrotron or portion of linac), then

$$tr M = 2 + \frac{2\pi h \eta}{\beta^2 E} QeV \cos\phi_s = 2\cos\Delta\psi_s \qquad \Delta\psi_s = 2\pi\nu_s$$

longitudinal phase advance

oscillation frequency
w.r.t. cavity number, "*n*"
(e.g., synchrotron *tune*)

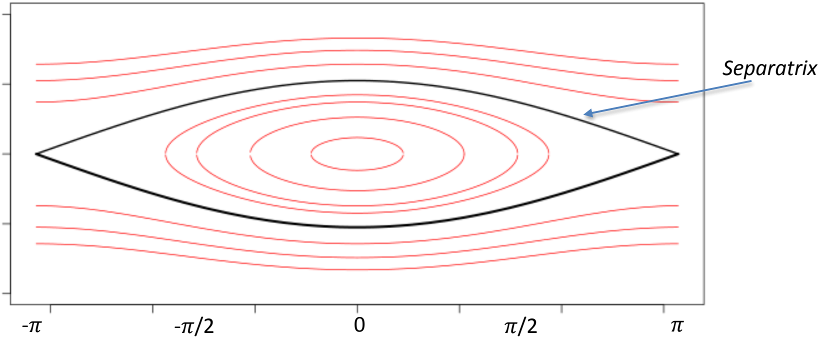
$$\cos\Delta\psi_s \approx 1 - \frac{1}{2}(\Delta\psi_s)^2 = 1 + \frac{\pi h \eta}{\beta^2 E} QeV \cos\phi_s \left[= \frac{1}{2} tr M \right]$$

$$\Rightarrow \qquad \nu_s = \sqrt{-\frac{h\eta}{2\pi\beta^2 E} QeV \cos\phi_s}$$

as found previously!

The Stationary Bucket

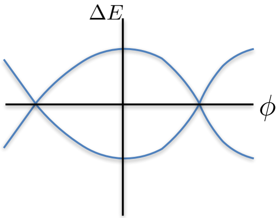
- Suppose do not wish to accelerate the ideal particle…
  - for lower energies, where the slip factor is negative, then need to choose $\phi_s = 0^o$



“stationary” bucket: $\phi_s = 0,\ 2\pi \qquad (\sin\phi_s = 0)$ —> no average acceleration

anticipate stability: —> choose $\phi_s = 0, \quad \eta < 0$

then, $$\Delta E^2 + 2\frac{\beta^2 E}{2\pi h\eta} QeV \cos\phi = constant$$



on the separatrix: $\Delta E = 0$ at $\phi = \pm\pi$

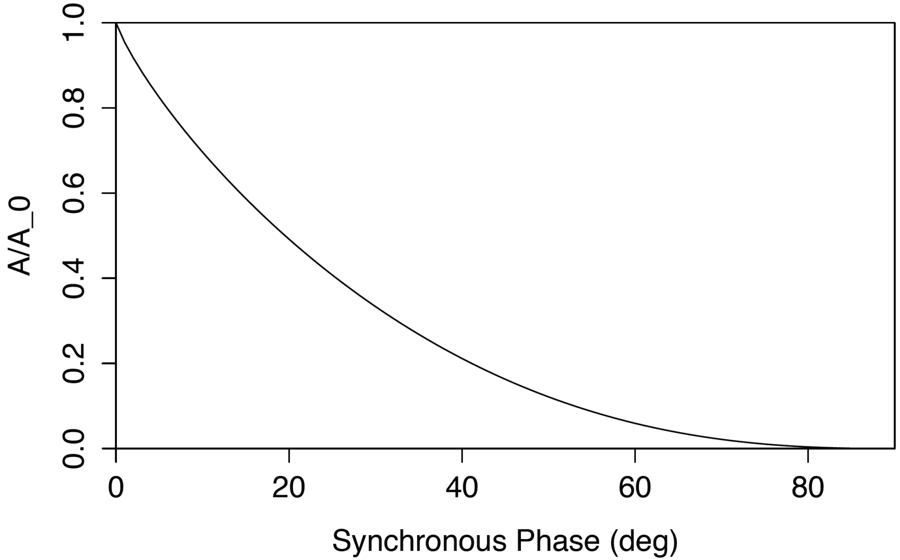
$$0 - 2\frac{\beta^2 E}{2\pi h\eta} QeV = constant$$

thus, the Eq. of separatrix: $$\Delta E^2 + (1 + \cos\phi)\frac{\beta^2 E}{\pi h\eta} QeV = 0$$

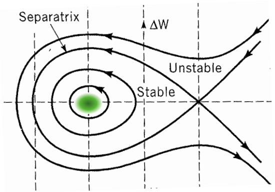
$$\Delta E^2 + \frac{2\beta^2 E}{\pi h\eta} QeV \cos^2(\phi/2) = 0$$

separatrix: $$\boxed{\Delta E = \pm\sqrt{-\frac{2\beta^2 E}{\pi h\eta} QeV} \cos(\phi/2)}$$ *(for “stationary bucket”)*

Numerical solution for bucket area



Back to Small Oscillations…



from (2), $\Delta E^2 + 2\frac{\beta^2 E}{2\pi h\eta} QeV(\cos\phi + \phi\sin\phi_s) = constant$

if $\phi = \phi_s + \Delta\phi$, then …

(small)

$$\Delta E^2 + 2\frac{\beta^2 E}{2\pi h\eta} QeV(\cos\phi_s\cos\Delta\phi - \sin\phi_s\sin\Delta\phi + (\phi_s + \Delta\phi)\sin\phi_s) = constant$$

$$\Delta E^2 + 2\frac{\beta^2 E}{2\pi h\eta} QeV(\cos\phi_s(1 - \frac{1}{2}\Delta\phi^2) - \sin\phi_s\Delta\phi$$
$$+\phi_s\sin\phi_s + \Delta\phi\sin\phi_s) = constant$$

$$\boxed{\Delta E^2 + \left(-\frac{\beta^2 E}{2\pi h\eta} QeV\cos\phi_s\right)\Delta\phi^2 = constant} \qquad (3)$$

This Eqn. represents trajectories in longitudinal phase space of particles ***near*** the ideal particle.

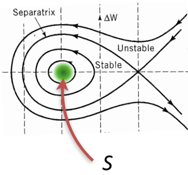
Beam Longitudinal Emittance

Suppose beam is well contained within an ellipse given by (3), and suppose we know either $\Delta\hat{E}$ or $\Delta\hat{\phi}$ (or, $\Delta\hat{t}$) of the distribution (i.e., maximum extent). Then, the *constant* is easily seen to be:

$$constant = \Delta\hat{E}^2 = -\frac{\beta^2 E}{2\pi h\eta} QeV\cos\phi_s \Delta\hat{\phi}^2$$

So, area of ellipse (the *longitudinal emittance*) is: $\pi\ \Delta\hat{E}\Delta\hat{\phi}$

or, in *E-t* coordinates, $S \equiv \pi\ \Delta\hat{E}\Delta\hat{t} = \pi\ \Delta\hat{E}\dfrac{\Delta\hat{\phi}}{2\pi f_{\rm rf}}$



$$\Rightarrow\quad S = \frac{1}{2f_{\rm rf}}\sqrt{-\frac{\beta^2 EeV}{2\pi h\eta}Q\cos\phi_S}\ \ \Delta\hat{\phi}^2$$

or,

$$\boxed{S = 2\pi^2 f_{\rm rf}\sqrt{-\frac{\beta^2 EeV}{2\pi h\eta}Q\cos\phi_S}\ \ \Delta\hat{t}^2}$$

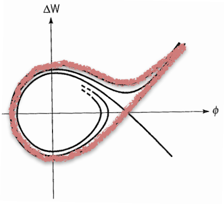
units: "eV-sec"

Golf Clubs *vs.* Fish

- Our analysis “assumes” slowly changing variables (including the energy gain!). Quite reasonable in many Alvarez-style linacs and in synchrotrons
- In linacs, fractional energy change can be large, and so this will distort the phase space
- Plots from Wangler’s book:

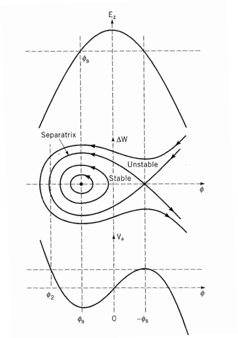
Here, a more rapid acceleration is included

(linac)



Here, assume that energy is “constant” or varying very slowly

(synchrotron)



Transition Energy

- In a synchrotron, there can be an energy at which the slip factor changes sign — this is call the "transition energy"

$$\eta = \alpha_p - \frac{1}{\gamma^2} = \left\langle \frac{D}{\rho} \right\rangle - \frac{1}{\gamma^2}$$

$$\eta = 0 = \alpha_p - \frac{1}{\gamma^2}$$

$$\eta = \frac{1}{\gamma_t^2} - \frac{1}{\gamma^2}$$

$$\boxed{\gamma_t \equiv \frac{1}{\sqrt{\alpha_p}}}$$

- In a typical FODO-style synchrotron, the transition gamma is roughly equal to the betatron tune

Transition

We had... $\Rightarrow \dfrac{d^2\Delta\phi}{dn^2} - \left(\dfrac{2\pi h\eta}{\beta^2 E} QeV\cos\phi_s\right)\Delta\phi = 0$

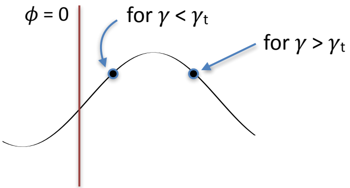
$$\nu_s = \sqrt{-\frac{h\eta}{2\pi\beta^2 E} QeV\cos\phi_s}$$

if $\eta > 0$, choose $\cos\phi_s < 0$

So,

when $\eta < 0$, we want $\cos\phi_s > 0$

when $\eta > 0$, we want $\cos\phi_s < 0$

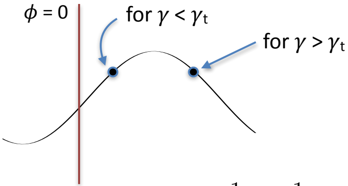


$\therefore$ if γ_t exists, need "phase jump" to occur at transition crossing

$\gamma_t mc^2 =$ transition energy

Transition Crossing

- If the synchrotron accelerates through its transition energy, then the phase of the RF system has to be shifted at the time of transition crossing
- The synchrotron motion slows down as approach transition — it would stop if the slip factor were exactly zero!
  - loss of phase stability!
  - momentum spread also gets larger near transition
- So, best to accelerate quickly through this energy region!



$$\eta = \frac{1}{\gamma_t^2} - \frac{1}{\gamma^2}$$

$$\boxed{\nu_s = \sqrt{-\frac{h\eta}{2\pi\beta^2 E} QeV \cos\phi_s}}$$

Buckets, Bunches, Batches, …

- Have seen definition of "buckets" — stable phase space area
- Buckets can be occupied by "bunches" of particles
  - note: need not be — can have "empty buckets"
  - thus, can (in principle) adjust bunch spacing, bunch arrangements, etc.
- A set of bunches that are created in an accelerator (pulsed) is often called a Batch (especially if from a synchrotron)
  - can also be called a Bunch Train as well (especially if from a linac)

Showing yet more animations